

# What Can These Data Support?

Unit 1 · Topic 1.1 · THE PROBLEM

Suppose a school newspaper records last night's sleep for 12 students in one class. All 12 values are valid. For this class, the school flags a sleep concern if more than 20% slept fewer than 6 hours.

**Claim 1:** "The full table shows a sleep concern in this class."

**Claim 2:** "Leave out the two lowest sleep values and there is no sleep concern."

Sleep last night

| Student | Sleep (hours) |
|---------|---------------|
| Aiden   | 5.0           |
| Bea     | 7.0           |
| Cruz    | 8.5           |
| Dev     | 6.5           |
| Elise   | 8.0           |
| Fiona   | 7.5           |
| Gabe    | 5.5           |
| Hana    | 9.0           |
| Imani   | 7.0           |
| Jalen   | 6.0           |
| Kira    | 8.0           |
| Leo     | 4.5           |

## FIRST TAKE — one sentence before you work the parts

Which claim, if either, would you print? Name one detail behind your choice.

**DOK 1** (a) Name the variable and its unit.

**DOK 2** (b) Find the percent who slept fewer than 6 hours using all 12 students. Then leave out the two lowest values and find the new percent.

**DOK 3** (c) Which claim would you print? Use both percents and the school's rule. Explain why keeping or removing the two values changes what readers learn about this class.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.